

# **COS1501**

# **STUDY UNIT 5**

# **RELATIONS**

## **Complete taught notes**

Ordered pairs • Cartesian products • relations • domain/range/codomain • properties • trichotomy • inverse relations • composition • proofs • worked examples

### **What this PDF contains**

Full source-aligned coverage of COS1501 Study Unit 5 (Relations). Every official concept is explained in lesson style, each topic has at least one worked example, the main source activities are worked through, and the final pages contain exam methods, common traps and a revision checklist.

### **Source alignment**

The terminology and structure follow the official UNISA COS1501 Study Unit 5 (pp. 69-82 in the study guide). Explanations, memory rules and added worked examples in this PDF are newly written for learning; they are not a verbatim reproduction of the guide.

## Unit 5 Roadmap

- 5.1 Ordered pairs: coordinates, order and why ordered pairs matter
- 5.2 Relations: arrow diagrams, infix notation, Cartesian products and relations as subsets
- Domain, range, codomain and binary relations
- 5.3 Properties: reflexive, irreflexive, symmetric, antisymmetric, transitive and trichotomy
- Building relations with required properties
- Inverse relations
- Composition of relations
- Worked source-style activities and the theorem  $R$  is transitive iff  $R \circ R \subseteq R$
- Exam methods, common traps, final checklist and practice

## Learning outcomes

- Distinguish a set from an ordered pair and read first/second coordinates correctly.
- Form  $A \times B$ ,  $B \times A$  and  $A^2$  and know why order matters.
- Decide whether a proposed set of pairs is a relation from  $A$  to  $B$ .
- Find the domain, range and codomain of a relation.
- Classify a relation as reflexive/irreflexive, symmetric/antisymmetric, transitive, and test trichotomy.
- Construct examples of relations with specified combinations of properties.
- Find  $R^{-1}$  and compute compositions such as  $S \circ R$ .
- Use composition to reason about transitivity and solve COS1501-style relation questions.

### Core idea

A relation is a set of ordered pairs. Once you know which ordered pairs are present, every property in Unit 5 is a precise rule about those pairs.

## 0. Big Picture: How the Unit Fits Together

Unit 5 starts with the smallest building block - an ordered pair - and then builds upward. A Cartesian product lists every allowed ordered pair between two sets. A relation chooses any subset of those allowed pairs. Properties such as symmetry or transitivity describe patterns inside the chosen subset. Inverse and composition then create new relations from old ones.

| Stage             | Meaning                                      | Example                                                |
|-------------------|----------------------------------------------|--------------------------------------------------------|
| Ordered pair      | Two coordinates where position matters       | $(2, 5)$                                               |
| Cartesian product | All possible allowed pairs                   | $A \times B$                                           |
| Relation          | Any subset of a Cartesian product            | $R \subseteq A \times B$                               |
| Relation on $A$   | Both coordinates come from $A$               | $R \subseteq A \times A = A^2$                         |
| Property          | A rule about which pairs must/must not occur | symmetric, transitive, ...                             |
| Inverse           | Reverse every ordered pair                   | $(x, y) \mapsto (y, x)$                                |
| Composition       | Join two compatible relation steps           | $a R b \text{ and } b S c \Rightarrow a (S \circ R) c$ |

# 1. Ordered Pairs (Official 5.1)

## Definition

An ordered pair has the form  $(x, y)$ .  $x$  is the first coordinate and  $y$  is the second coordinate. Unlike a set, the positions matter: in general  $(x, y) \neq (y, x)$ .

Compare the notation. In a set,  $\{1, 2\} = \{2, 1\}$ ; the order used to list set elements is irrelevant. In an ordered pair,  $(3, 0) \neq (0, 3)$  because the first and second positions carry different information.

## 1.1 Ordered pairs on the Cartesian plane

The historical geometric use of ordered pairs is to locate points. The first coordinate tells you how far to move along the horizontal  $x$ -axis; the second tells you how far to move along the vertical  $y$ -axis. The origin is  $(0, 0)$ . Positive  $x$  is right, negative  $x$  is left; positive  $y$  is up, negative  $y$  is down.

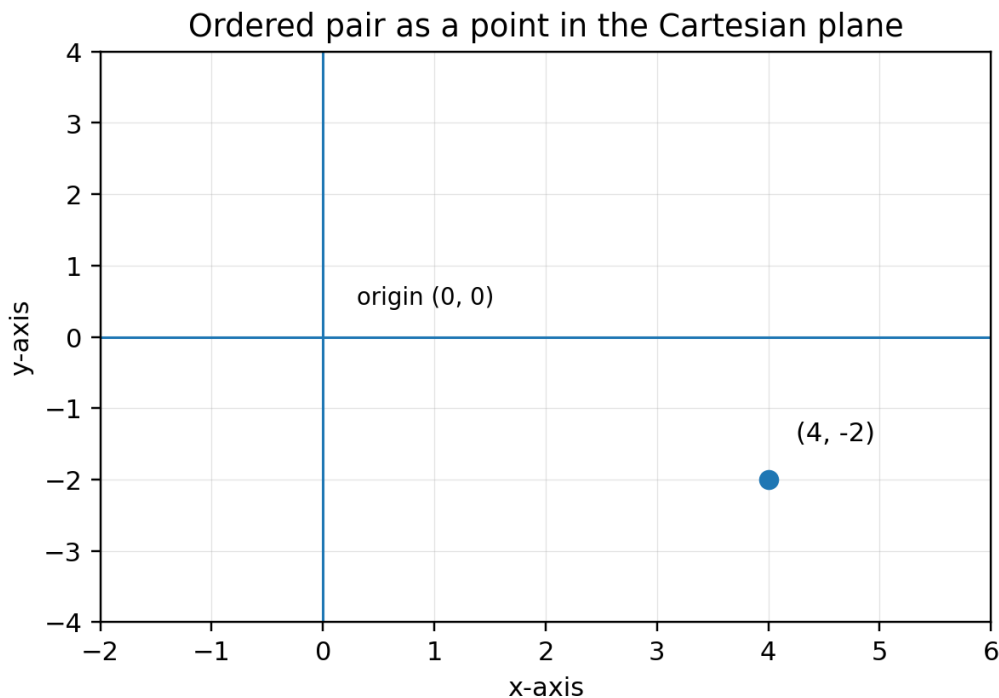


Figure 1. The source example  $(4, -2)$ : move 4 units right and 2 units down.

## Worked example - read an ordered pair

Point  $P = (-3, 2)$ . The first coordinate  $-3$  means move 3 units left from the origin.

The second coordinate  $2$  means move 2 units upward.

So  $(-3, 2)$  and  $(2, -3)$  are different points because their coordinate roles are reversed.

## 1.2 Why ordered pairs are useful beyond graphs

The fixed coordinate positions let an ordered pair encode a relationship. For example,  $(\text{student}, \text{module})$  can mean “this student is registered for this module”;  $(\text{symbolic\_name}, \text{memory\_location})$  can mean “this program name is stored at this address”. A set of such ordered pairs is a relation.

## Added example - relationship information

Let  $(\text{Amina}, \text{COS1501})$  mean “Amina studies COS1501”.

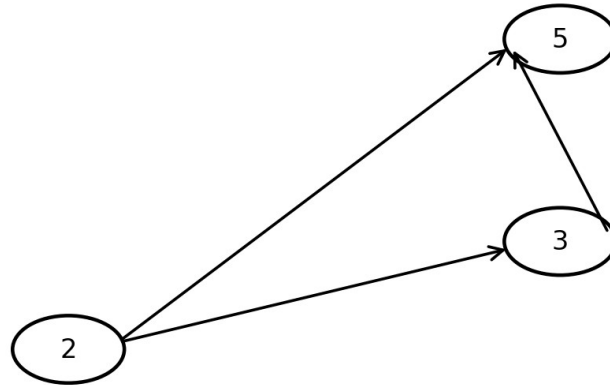
The reversed pair  $(\text{COS1501}, \text{Amina})$  would mean something different because the first and second coordinate roles have changed.

## 2. Relations (Official 5.2)

### 2.1 Relation represented by arrows and ordered pairs

Consider  $\{2, 3, 5\}$  and the relationship “is less than”. The true comparisons are  $2 < 3$ ,  $2 < 5$  and  $3 < 5$ . We can draw arrows for those comparisons or store the same information as ordered pairs.

Relation “is less than” on  $\{2, 3, 5\}$



$R = \{(2,3), (2,5), (3,5)\}$

#### Key representation

$R = \{(2, 3), (2, 5), (3, 5)\}$ . This set of ordered pairs completely records the arrows for the “<” relation on  $\{2,3,5\}$ .

### 2.2 Infix notation

Instead of writing  $(x, y) \in R$ , we may write  $x R y$ . This is called infix notation because the relation symbol is written between the two elements. If the relation itself is “<”, then  $(2, 3) \in <$  is normally written simply as  $2 < 3$ .

#### Worked example - infix notation

Let  $R = \{(a,b),(b,c)\}$ .

$(a,b) \in R$  can be written  $a R b$ .

$(c,a) \notin R$  can be written “c is not R-related to a”.

### 2.3 Relations can connect different kinds of objects

A relation need not connect numbers to numbers. The source illustrates a relation from a set  $M$  of men to a set  $W$  of women, where  $(x_i, y_i)$  records “ $x_i$  is the husband of  $y_i$ ”. The main point is not the real-life example itself; it is that the first coordinates may come from one set and the second coordinates from another set.

### 2.4 Computing connection: symbol tables

The study guide also connects relations to a compiler symbol table. Let  $S$  be symbolic names,  $A$  be possible attributes/types, and  $L$  be memory locations. One relation can pair a symbol in  $S$  with an attribute in  $A$ , while another pairs a symbol in  $S$  with a location in  $L$ . This is exactly the kind of structured association relations model.

#### Added computing example

Let  $S = \{\text{score, passed}\}$ ,  $A = \{\text{integer, Boolean}\}$ .

$R = \{(\text{score, integer}), (\text{passed, Boolean})\}$  is a relation from  $S$  to  $A$ .

The first coordinate is a program name; the second coordinate is its type.

### 3. Cartesian Products

#### Definition

For sets  $A$  and  $B$ ,  $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$ . Every element of  $A$  is paired with every element of  $B$ .

$A \times B$  is the universe of possible ordered pairs for a relation from  $A$  to  $B$ . A relation from  $A$  to  $B$  is allowed to contain any of these pairs, including none of them or all of them.

#### Official-style example

$A = \{2, 3, 4\}$ ,  $B = \{5, 6\}$ .

$A \times B = \{(2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$ .

$B \times A = \{(5, 2), (5, 3), (5, 4), (6, 2), (6, 3), (6, 4)\}$ .

$B^2 = B \times B = \{(5, 5), (5, 6), (6, 5), (6, 6)\}$ .

$A^2 = A \times A$  contains  $3 \times 3 = 9$  ordered pairs.

#### Exam trap

Cartesian product is generally not commutative:  $A \times B$  usually differs from  $B \times A$  because reversing the set order reverses coordinate roles.

#### 3.1 Fast method and size check

If  $|A| = m$  and  $|B| = n$  for finite sets, then  $|A \times B| = mn$ . Use this as a checking tool. If  $A$  has 4 elements and  $B$  has 2,  $A \times B$  must contain 8 ordered pairs.

#### Worked Activity 5-4

$A = \{1, 2, 3, 4\}$ ,  $B = \{2, 5\}$ ,  $C = \{3, 4, 7\}$ .

(a)  $A \times B = \{(1, 2), (1, 5), (2, 2), (2, 5), (3, 2), (3, 5), (4, 2), (4, 5)\}$ .

(b)  $B \times A = \{(2, 1), (2, 2), (2, 3), (2, 4), (5, 1), (5, 2), (5, 3), (5, 4)\}$ .

(c)  $A \cup B = \{1, 2, 3, 4, 5\}$ , so  $(A \cup B) \times C$  has 15 pairs: each of 1, 2, 3, 4, 5 paired with 3, 4, 7.

(d) Using  $+$  for symmetric difference,  $A + B = \{1, 3, 4, 5\}$ ; hence  $(A + B) \times B = \{(1, 2), (1, 5), (3, 2), (3, 5), (4, 2), (4, 5), (5, 2), (5, 5)\}$ .

### 4. A Relation as a Subset of a Cartesian Product

#### Definition

A subset of  $C \times D$  is called a relation from  $C$  to  $D$ . So  $R$  is a relation from  $C$  to  $D$  exactly when  $R \subseteq C \times D$ .

Because the empty set is a subset of every set,  $\emptyset$  is also a valid relation from  $C$  to  $D$ . The full Cartesian product  $C \times D$  is another valid relation. Every other relation lies somewhere between those extremes.

#### Worked example - valid relations

Let  $A = \{2, 3, 4\}$ ,  $B = \{6, 7\}$ .

$\emptyset$ ,  $\{(3, 7)\}$ ,  $\{(2, 6), (2, 7)\}$ , and  $A \times B$  are all relations from  $A$  to  $B$ .

$\{(6, 2)\}$  is not a relation from  $A$  to  $B$  because 6 is not allowed as a first coordinate and 2 is not allowed as a second coordinate.

#### 4.1 Domain, range and codomain

#### Definitions

If  $T$  is a relation from  $X$  to  $Y$ , then  $Y$  is the codomain.  $\text{dom}(T)$  is the set of first coordinates that actually occur.  $\text{ran}(T)$  is the set of second coordinates that actually occur. Thus  $\text{dom}(T) \subseteq X$  and  $\text{ran}(T) \subseteq Y$ .

#### Worked example

Let  $S = \{(a, 1), (b, 1), (a, 2)\}$  be a relation from  $X = \{a, b, c\}$  to  $Y = \{1, 2, 3\}$ .

$\text{dom}(S) = \{a, b\}$ : these are the first coordinates actually used.

$\text{ran}(S) = \{1, 2\}$ : these are the second coordinates actually reached.

$\text{codomain}(S)=Y=\{1,2,3\}$ . Notice that  $c$  is unused in the domain and  $3$  is unused in the range.

### Range vs codomain

Do not assume  $\text{range} = \text{codomain}$ . The codomain is the entire target set declared for the relation; the range is only what actually appears as a second coordinate.

## 4.2 Binary relations and relations on a set

Any  $R \subseteq X \times Y$  is called a binary relation from  $X$  to  $Y$  because its members are ordered pairs with two coordinates. If both coordinate sets are  $X$ , so  $R \subseteq X \times X$ , then  $R$  is a binary relation on  $X$ . The notation  $R \subseteq A^2$  means exactly  $R \subseteq A \times A$ :  $R$  is a relation on  $A$ .

### Added example - relation on A

$A=\{1,2,3\}$ ;  $R=\{(1,1),(1,3),(2,2)\}$ .

Every first and second coordinate belongs to  $A$ , so  $R \subseteq A^2$  and  $R$  is a relation on  $A$ .

## 5. Properties of Relations (Official 5.3)

From this point the guide mainly studies relations on one set  $A$ , i.e.  $R \subseteq A \times A$ . Each property asks a different question. Do not merge the definitions: a relation can have several properties at once, or fail several at once.

| Property      | What to check                          | Fast memory                                    |
|---------------|----------------------------------------|------------------------------------------------|
| Reflexive     | Every $(x,x)$ is present               | all self-loops                                 |
| Irreflexive   | No $(x,x)$ is present                  | no self-loops                                  |
| Symmetric     | Every $(x,y)$ has $(y,x)$              | mirror must exist                              |
| Antisymmetric | No distinct $x,y$ have both directions | no two-way pair between distinct elements      |
| Transitive    | $(x,y)$ and $(y,z)$ force $(x,z)$      | two steps force shortcut                       |
| Trichotomy    | Every distinct $x,y$ are comparable    | at least one direction for every distinct pair |

## 6. Reflexive Relations

### Definition

$R$  on  $A$  is reflexive on  $A$  iff for every  $x \in A$ ,  $(x, x) \in R$ .

The set  $A$  determines exactly which diagonal pairs are required. Extra non-diagonal pairs do not destroy reflexivity.

### Worked example - finite set

$A=\{2,3,5\}$ . The required diagonal is  $\{(2,2),(3,3),(5,5)\}$ .

$S=\{(2,2),(3,3),(5,5),(2,3)\}$  is reflexive because all three required self-pairs are present.

### Worked example - rule on integers

Define  $(x,y) \in R$  iff  $x-y=7k$  for some  $k \in \mathbb{Z}$ .

To test reflexivity set  $y=x$ :  $x-x=0=7 \cdot 0$ .

$k=0$  is an integer, so  $(x,x) \in R$  for every integer  $x$ . Therefore  $R$  is reflexive.

### How to disprove reflexivity

Find one  $x \in A$  for which  $(x,x)$  is missing. One missing required diagonal pair is enough.

## 7. Irreflexive Relations

### Definition

$R$  on  $A$  is irreflexive iff for every  $x \in A$ ,  $(x, x) \notin R$ . No element may be related to itself.

### Worked example

$A = \{2, 3, 5\}$ ;  $R = \{(3, 2), (2, 5), (3, 5)\}$ .

None of  $(2, 2), (3, 3), (5, 5)$  appears, so  $R$  is irreflexive.

### Rule example

On  $\mathbb{Z}$ , let  $x R y$  mean  $x < y$ .

No integer satisfies  $x < x$ , so no  $(x, x)$  belongs to  $R$ . Hence  $<$  is irreflexive.

## 7.1 A relation can be neither reflexive nor irreflexive

These are not simple opposites. Reflexive requires every diagonal pair; irreflexive requires none. If only some diagonal pairs occur, the relation is neither.

### Worked example - neither

$A = \{2, 3, 5\}$ ;  $S = \{(2, 2), (2, 5), (3, 5)\}$ .

$S$  is not reflexive because  $(3, 3)$  and  $(5, 5)$  are missing.

$S$  is not irreflexive because  $(2, 2)$  is present.

Therefore  $S$  is neither.

## 8. Symmetric Relations

### Definition

$R$  on  $A$  is symmetric iff for all  $x, y \in A$ ,  $(x, y) \in R$  implies  $(y, x) \in R$ .

Think mirror image across the ordered-pair diagonal: every arrow  $x \rightarrow y$  must have a reverse arrow  $y \rightarrow x$ . Diagonal pairs  $(x, x)$  are automatically their own mirror.

### Source-style finite examples

$R_1 = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$  is symmetric and irreflexive.

$R_2 = \{(1, 1), (2, 2), (3, 3), (2, 3)\}$  is reflexive but not symmetric because  $(3, 2)$  is missing.

$R_3 = \{(1, 1), (2, 2), (3, 3), (2, 3), (3, 2)\}$  is both reflexive and symmetric.

### Rule example - congruence modulo 7

Assume  $x - y = 7k$ , so  $(x, y) \in R$ .

Then  $y - x = -(x - y) = 7(-k)$ , and  $-k \in \mathbb{Z}$ .

Therefore  $(y, x) \in R$ . The relation is symmetric.

### How to disprove symmetry

One counterexample is enough: show  $(x, y) \in R$  but  $(y, x) \notin R$ .

## 9. Antisymmetric Relations

### Definition

$R$  on  $A$  is antisymmetric iff for all  $x, y \in A$ , if  $x \neq y$  and  $(x, y) \in R$ , then  $(y, x) \notin R$ . Equivalent form: if  $(x, y) \in R$  and  $(y, x) \in R$ , then  $x = y$ .

Antisymmetric does NOT mean “not symmetric”. The restriction only targets two-way pairs between different elements. Self-pairs such as  $(b, b)$  are completely allowed.

### Worked example

$A = \{a, b, c\}$ ;  $P = \{(a, b), (b, b), (b, c)\}$ .

$(a, b)$  occurs but  $(b, a)$  does not;  $(b, c)$  occurs but  $(c, b)$  does not.

The self-pair  $(b, b)$  is allowed. Hence  $P$  is antisymmetric.

**Rule example -  $\leq$  on integers**

If  $x \leq y$  and  $y \leq x$ , the only possibility is  $x = y$ .  
Therefore  $\leq$  is antisymmetric.

**9.1 Activity 5-6: “not symmetric” is not antisymmetric****Counterexample from the source activity**

$R = \{(1,2), (2,1), (2,3)\}$  on  $\{1,2,3\}$ .

$R$  is not symmetric because  $(2,3)$  is present but  $(3,2)$  is absent.

$R$  is also not antisymmetric because  $(1,2)$  and  $(2,1)$  are both present while  $1 \neq 2$ .

So “not symmetric” and “antisymmetric” are different ideas.

**Fast contrast**

Symmetric asks for reverse pairs. Antisymmetric forbids reverse pairs for distinct elements. A relation can even be both symmetric and antisymmetric when its only pairs are diagonal pairs.

**10. Transitive Relations****Definition**

$R$  on  $A$  is transitive iff for all  $x, y, z \in A$ , whenever  $(x, y) \in R$  and  $(y, z) \in R$ , then  $(x, z) \in R$ .

The middle coordinate must match. If you can travel  $x \rightarrow y$  and then  $y \rightarrow z$ , a transitive relation requires the shortcut  $x \rightarrow z$ . If there is no two-step chain in the relation, there is no transitivity requirement triggered by that chain.

**Worked example - finite relation**

$R = \{(1,1), (2,2), (1,2), (2,1)\}$  on  $\{1,2,3\}$ .

$(1,2)$  and  $(2,1)$  force  $(1,1)$ , which is present.

$(2,1)$  and  $(1,2)$  force  $(2,2)$ , which is present.

Chains involving the self-pairs also lead to pairs already in  $R$ . Therefore  $R$  is transitive.

**Rule example - congruence modulo 7**

Assume  $x - y = 7k$  and  $y - z = 7m$ .

Add them:  $x - z = 7k + 7m = 7(k + m)$ .

$k + m \in \mathbb{Z}$ , so  $(x, z) \in R$ . Thus the relation is transitive.

**Non-example - sports results**

Even if Team A beats Team B and Team B beats Team C, it does not follow that Team A must beat Team C. So “beats” in a competition is generally not a transitive relation.

**How to disprove transitivity**

Find a broken chain:  $(x, y)$  and  $(y, z)$  are in  $R$  but the forced pair  $(x, z)$  is missing.

**11. Trichotomy****Definition**

$R$  on  $A$  satisfies trichotomy iff every two distinct elements  $x, y \in A$  are comparable:  $xRy$  or  $yRx$ , i.e.  $(x, y) \in R$  or  $(y, x) \in R$ . The “or” is inclusive in the guide, so both directions may occur.

Trichotomy is about coverage of every distinct pair of elements. Diagonal pairs do not decide it because the condition applies when  $x \neq y$ .

**Worked example - satisfies trichotomy**

$A = \{1, 2, 3\}$ . Suppose  $S$  contains  $(3,2), (2,1), (3,1)$  (and it may contain diagonal pairs too).

Distinct pairs  $\{1,2\}$ ,  $\{1,3\}$ ,  $\{2,3\}$  are all comparable in at least one direction.

So  $S$  satisfies trichotomy.

### Worked non-example

$T = \{(3,2), (2,1), (2,3)\}$  on  $\{1,2,3\}$ .

For 1 and 3, neither  $(1,3)$  nor  $(3,1)$  is present.

Therefore 1 and 3 are not comparable and trichotomy fails.

### Rule example - $\leq$ on $\mathbb{Z}$

For distinct integers  $x$  and  $y$ , either  $x < y$  or  $y < x$ ; therefore either  $x \leq y$  or  $y \leq x$ .

Hence  $\leq$  satisfies trichotomy.

### Exam method

For a finite set with  $n$  elements, list each unordered pair of distinct elements once. For  $A = \{1,2,3,4\}$ , check  $\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}$ . Each must have at least one direction in  $R$ .

## 12. Building Relations with Required Properties

COS1501 can ask you to construct a relation with several properties at the same time. The safest method is to add only what each definition forces, then re-check all earlier properties after every addition.

### 12.1 Source Activity 5-7: reflexive + symmetric + transitive

#### Closure construction on $A = \{1,2,3\}$

Start reflexive: include  $(1,1), (2,2), (3,3)$ .

Suppose we also include  $(1,2)$  and  $(2,3)$ . Symmetry forces  $(2,1)$  and  $(3,2)$ .

Transitivity from  $(1,2)$  and  $(2,3)$  forces  $(1,3)$ .

Symmetry then forces  $(3,1)$ .

At that point every pair in  $A \times A$  is present. So  $R = A \times A$  is reflexive, symmetric and transitive.

### 12.2 Minimal alternatives

The source activity illustrates how adding some off-diagonal pairs can force many more. However, if a question only asks for any example, the identity relation  $I_A = \{(x,x) | x \in A\}$  is often the simplest relation that is reflexive, symmetric and transitive.

#### Added example - identity relation

$A = \{a,b,c\}$ .  $I_A = \{(a,a), (b,b), (c,c)\}$ .

Reflexive: every required diagonal pair exists.

Symmetric: each pair is its own reverse.

Transitive: any possible two-step chain stays on the same element.

## 13. Inverse Relations

### Definition

Given  $R$ , the inverse relation  $R^{-1}$  is obtained by reversing every ordered pair:  $(x,y) \in R$  iff  $(y,x) \in R^{-1}$ .

If  $R$  has domain  $A$  and range  $B$ , then  $R^{-1}$  reverses those roles: its domain is  $B$  and its range is  $A$ . Also,  $(R^{-1})^{-1} = R$ .

### Official example

$R = \{(a,b), (b,c), (a,c)\}$ .

Reverse every pair:

$R^{-1} = \{(b,a), (c,b), (c,a)\}$ .

### Added example

$R = \{(1,x), (2,y), (3,x)\}$ .

$$R^{-1} = \{(x,1), (y,2), (x,3)\}.$$

Notice one first coordinate  $x$  in  $R^{-1}$  can appear with more than one second coordinate; an inverse relation need not be a function.

### Fast method

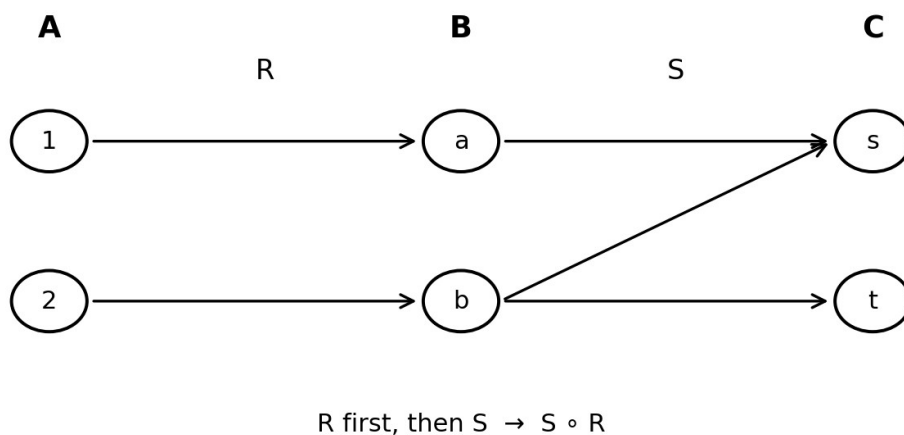
Do not rearrange the set or “solve” anything. For an explicitly listed relation, just flip every pair coordinate-by-coordinate.

## 14. Composition of Relations

### Definition

If  $R$  is from  $A$  to  $B$  and  $S$  is from  $B$  to  $C$ , then the composition of  $R$  followed by  $S$  is  $S \circ R$  (also written  $R;S$ ):  
 $S \circ R = \{(a,c) \mid \text{there exists } b \in B \text{ with } (a,b) \in R \text{ and } (b,c) \in S\}.$

The notation is easy to reverse accidentally. Read  $S \circ R$  from right to left:  $R$  happens first, then  $S$ . The semicolon notation  $R;S$  is included in the guide precisely because it visually reads in the order of travel.



### Official-style composition example

$R = \{(1,a), (2,b)\}$  and  $S = \{(a,s), (b,s), (b,t)\}.$

1  $R$  a and a  $S$  s, so  $(1,s) \in S \circ R.$

2  $R$  b and b  $S$  s/t, so  $(2,s), (2,t) \in S \circ R.$

Therefore  $S \circ R = \{(1,s), (2,s), (2,t)\}.$

### 14.1 Composition is generally order-sensitive

In the same example,  $R \circ S$  means  $S$  first then  $R$ . But  $S$  outputs  $s$  or  $t$ , while  $R$  expects  $a$  or  $b$  as first coordinates. No links match, so  $R \circ S = \emptyset$ . This shows that relation composition is generally not commutative.

### Matching rule

To compose, match the SECOND coordinate of a pair from the first relation with the FIRST coordinate of a pair from the second relation. Keep the outside coordinates.

### Added composition example

$R = \{(1,a), (1,b), (2,b)\}; S = \{(a,5), (b,6)\}.$

$1 \rightarrow a \rightarrow 5$  gives  $(1,5); 1 \rightarrow b \rightarrow 6$  gives  $(1,6); 2 \rightarrow b \rightarrow 6$  gives  $(2,6).$

$S \circ R = \{(1,5), (1,6), (2,6)\}.$

## 15. Self-Composition and Inverse/Composition Combinations

The source uses small relations on  $X = \{a,b,c\}$  to show that  $R \circ R$ ,  $R^{-1} \circ R$  and  $R \circ R^{-1}$  can all be different. These are excellent exercises because the notation is the main difficulty, not arithmetic.

**Example A - relation equal to its inverse**

$$R = \{(a,a), (b,c), (c,b)\}.$$

Reversing the pairs gives exactly the same set, so  $R^{-1} = R$ .

$$R \circ R = \{(a,a), (b,b), (c,c)\} \text{ because } a \rightarrow a \rightarrow a, b \rightarrow c \rightarrow b \text{ and } c \rightarrow b \rightarrow c.$$

**Example B -  $R = \{(a,b), (b,c), (a,c)\}$** 

$$R^{-1} = \{(b,a), (c,b), (c,a)\}.$$

$R \circ R = \{(a,c)\}$  because only  $a \rightarrow b \rightarrow c$  makes a two-step R-chain.

$$R^{-1} \circ R = R; R^{-1} = \{(a,a), (b,a), (b,b), (a,b)\}.$$

$$R \circ R^{-1} = R^{-1}; R = \{(b,b), (b,c), (c,c), (c,b)\}.$$

$$R^{-1} \circ R^{-1} = \{(c,a)\}.$$

**Notation check**

When in doubt, ignore the symbols for a moment and write “first \_\_\_\_, then \_\_\_\_”. For  $S \circ R$ , write “first R, then S”.

## 16. Master Exam Method: Classify a Finite Relation

When a question gives A and an explicit relation R, classify it in a fixed order. This prevents you from mixing definitions.

1. Write the required diagonal  $D = \{(x,x) | x \in A\}$ .
2. Reflexive? Check  $D \subseteq R$ . Irreflexive? Check  $D \cap R = \emptyset$ . If some but not all diagonals occur, it is neither.
3. Symmetric? For every off-diagonal  $(x,y)$  in R, search for  $(y,x)$ . One missing reverse disproves it.
4. Antisymmetric? Search for any distinct  $x,y$  with both  $(x,y)$  and  $(y,x)$ . One such pair disproves it.
5. Transitive? Build every possible two-step chain  $(x,y), (y,z)$ . Each must have  $(x,z)$ . One missing shortcut disproves it.
6. Trichotomy? Check every unordered pair of distinct elements; at least one direction must occur.

**Integrated classification example**

$$A = \{1, 2, 3\}; R = \{(1,1), (2,2), (3,3), (1,2), (2,3), (1,3)\}.$$

Reflexive: yes - all diagonal pairs are present.

Irreflexive: no - diagonal pairs exist.

Symmetric: no -  $(1,2)$  exists but  $(2,1)$  does not.

Antisymmetric: yes - no distinct pair occurs in both directions.

Transitive: yes - the important chain  $(1,2), (2,3)$  has shortcut  $(1,3)$ , and all chains involving self-pairs close correctly.

Trichotomy: yes -  $1/2$ ,  $1/3$  and  $2/3$  are each comparable.

## 17. Full Worked Source-Style Activity 5-8

### 17.1 Question 1: classify and compose a relation

Let  $A = \{1, 2, 3, \{1\}, \{2\}\}$ . Let  $P = \{(1, \{1\}), (1, 2)\}$  and  $R = \{(1, \{1\}), (1, 3), (2, \{1\}), (2, \{2\}), (\{1\}, 3), (\{2\}, \{1\})\}$ .

Remember that 1 and  $\{1\}$  are different objects, as are 2 and  $\{2\}$ .

| Part               | Answer | Reason                                                                                               |
|--------------------|--------|------------------------------------------------------------------------------------------------------|
| (a) Irreflexive?   | Yes    | No pair has identical first and second coordinates. $(1, \{1\})$ is not a diagonal pair.             |
| (b) Reflexive?     | No     | Pairs such as $(1,1)$ , $(2,2)$ , $(3,3)$ , $(\{1\}, \{1\})$ , $(\{2\}, \{2\})$ are not all present. |
| (c) Symmetric?     | No     | $(1, \{1\}) \in R$ but $(\{1\}, 1) \notin R$ .                                                       |
| (d) Antisymmetric? | Yes    | There is no two-way pair between two distinct elements.                                              |
| (e) Transitive?    | No     | $(2, \{1\})$ and $(\{1\}, 3)$ are in R, but $(2, 3)$                                                 |

| Part                                                                                                                                                                                                                                                                                                                                                                        | Answer | Reason                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                             |        | is not.                                                   |
| (f) Trichotomy?                                                                                                                                                                                                                                                                                                                                                             | No     | 1 and 2 are distinct but neither (1,2) nor (2,1) is in R. |
| <b>(g) Compute <math>R \circ R</math></b><br>Find all chains $x R y$ and $y R z$ .<br>$1 \rightarrow \{1\} \rightarrow 3$ gives (1,3).<br>$2 \rightarrow \{1\} \rightarrow 3$ gives (2,3).<br>$2 \rightarrow \{2\} \rightarrow \{1\}$ gives (2,{1}).<br>$\{2\} \rightarrow \{1\} \rightarrow 3$ gives ({2},3).<br>So $R \circ R = \{(1,3), (2,3), (2,\{1\}), (\{2\},3)\}$ . |        |                                                           |
| <b>(h) Compute <math>R \circ P</math></b><br>$R \circ P$ means P first, then R.<br>From P: $1 \rightarrow \{1\}$ , and $\{1\} \rightarrow 3$ under R, giving (1,3).<br>From P: $1 \rightarrow 2$ , and $2 \rightarrow \{1\}$ or $2 \rightarrow \{2\}$ under R, giving (1,{1}) and (1,{2}).<br>Therefore $R \circ P = \{(1,3), (1,\{1\}), (1,\{2\})\}$ .                     |        |                                                           |
| <b>(i) <math>T = \{(a,B) \in R \mid a \in B\}</math></b><br>Check only pairs whose second coordinate is a set.<br>(1,{1}) qualifies because $1 \in \{1\}$ .<br>(2,{2}) qualifies because $2 \in \{2\}$ .<br>(2,{1}) does not qualify; ({2},{1}) does not qualify because {2} is not an element of {1}.<br>$T = \{(1,\{1\}), (2,\{2\})\}$ .                                  |        |                                                           |

## 17.2 Question 2: construct relations on $P(\{a,b\})$

Let  $U = P(\{a,b\}) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$ . The following are valid examples; many other answers are possible.

| Requirement                                                    | Example relation                                                                                                                | Why it works                                                                                                                                                   |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) Reflexive, symmetric, transitive                           | Identity on U: $\{(X,X) \mid X \in U\}$                                                                                         | All diagonals; every pair mirrors itself; all chains stay at same X.                                                                                           |
| (b) Reflexive, symmetric, not transitive                       | All four diagonals + $\emptyset \leftrightarrow \{a\} + \{a\} \leftrightarrow \{b\}$ , but no $\emptyset \leftrightarrow \{b\}$ | The chain $\emptyset \rightarrow \{a\} \rightarrow \{b\}$ exists while $\emptyset \rightarrow \{b\}$ is missing.                                               |
| (c) Reflexive, transitive, neither symmetric nor antisymmetric | All diagonals + $\emptyset \leftrightarrow \{a\} + \emptyset \rightarrow \{b\} + \{a\} \rightarrow \{b\}$                       | Not symmetric because $\{b\} \rightarrow \emptyset$ is absent; not antisymmetric because $\emptyset \leftrightarrow \{a\}$ ; transitive closures are included. |
| (d) Symmetric and antisymmetric                                | Identity relation                                                                                                               | Only diagonal pairs occur, so both properties hold.                                                                                                            |
| (e) Irreflexive, antisymmetric, transitive                     | $\emptyset \rightarrow \{a\}, \{a\} \rightarrow \{b\}, \emptyset \rightarrow \{b\}$                                             | No diagonals, no reverse distinct pairs, and the required shortcut $\emptyset \rightarrow \{b\}$ is present.                                                   |

## 17.3 Question 3: theorem connecting transitivity and composition

### Theorem

For a relation R on X: R is transitive if and only if  $R \circ R \subseteq R$ .

### Proof direction 1: transitive $\Rightarrow R \circ R \subseteq R$

1. Assume R is transitive.
2. Let  $(x,z) \in R \circ R$ .
3. By the definition of composition, there exists  $y \in X$  such that  $(x,y) \in R$  and  $(y,z) \in R$ .
4. Because R is transitive,  $(x,z) \in R$ .
5. Therefore every element of  $R \circ R$  belongs to R, so  $R \circ R \subseteq R$ .

**Proof direction 2:  $R \circ R \subseteq R \Rightarrow$  transitive**

1. Assume  $R \circ R \subseteq R$ .
2. Suppose  $(x,y) \in R$  and  $(y,z) \in R$ .
3. By the definition of composition,  $(x,z) \in R \circ R$ .
4. Because  $R \circ R \subseteq R$ , it follows that  $(x,z) \in R$ .
5. Hence whenever  $(x,y)$  and  $(y,z)$  are in  $R$ ,  $(x,z)$  is in  $R$ . Therefore  $R$  is transitive.

**Why this theorem matters**

$R \circ R$  lists every two-step  $R$ -path as one direct ordered pair. Transitivity says every such two-step result must already be in  $R$ . That is exactly the subset condition  $R \circ R \subseteq R$ .

**18. Source Activity Coverage Checklist**

| Source activity | What it tests                                             | Covered here                                        |
|-----------------|-----------------------------------------------------------|-----------------------------------------------------|
| 5-1             | Mind map / overview                                       | Roadmap and big-picture section                     |
| 5-2             | Concept vocabulary                                        | Definitions throughout + final glossary table       |
| 5-3             | Examples of relations from A to B                         | Sections 2-4                                        |
| 5-4             | Cartesian products                                        | Worked in Section 3.1                               |
| 5-5             | CAI tutorial reminder                                     | All listed concepts are taught directly in this PDF |
| 5-6             | Symmetry vs antisymmetry                                  | Section 9.1                                         |
| 5-7             | Build reflexive, symmetric, transitive relation           | Section 12.1                                        |
| 5-8             | Mixed classification, construction, composition and proof | Section 17                                          |

**19. Common Exam Traps**

- Confusing  $\{a,b\}$  with  $(a,b)$ . Braces make a set; parentheses make an ordered pair.
- Assuming  $A \times B = B \times A$ . The coordinates come from different positions; order matters.
- Calling the declared source set the domain. In this guide,  $\text{dom}(R)$  is the first coordinates that actually appear.
- Calling range and codomain the same automatically.  $\text{ran}(R)$  may be a proper subset of the codomain.
- Testing reflexivity without using the whole underlying set  $A$ . You must know which diagonal pairs are required.
- Thinking “not reflexive” means irreflexive. A relation with some but not all self-pairs is neither.
- Thinking “not symmetric” means antisymmetric. It does not.
- Thinking antisymmetric forbids self-pairs. It only forbids two-way pairs for distinct elements.
- Testing transitivity by looking for three different elements only.  $x,y,z$  are allowed to repeat.
- Assuming transitivity fails when a two-step chain does not exist. If the antecedent never occurs, that chain causes no violation.
- Testing trichotomy using only pairs already in  $R$ . You must consider every distinct pair from  $A$ .
- Reading  $S \circ R$  left-to-right.  $R$  happens first, then  $S$ .
- For composition, matching first-to-first instead of second coordinate of the first relation to first coordinate of the second.

**20. Final Quick-Reference Sheet**

| Concept              | Definition / formula                                                 |
|----------------------|----------------------------------------------------------------------|
| Ordered pair         | $(x,y)$ : $x$ first coordinate, $y$ second coordinate; order matters |
| Cartesian product    | $A \times B = \{(x,y)   x \in A \text{ and } y \in B\}$              |
| Relation from A to B | $R \subseteq A \times B$                                             |
| Relation on A        | $R \subseteq A^2 = A \times A$                                       |

| Concept              | Definition / formula                                                         |
|----------------------|------------------------------------------------------------------------------|
| Domain               | $\text{dom}(R) = \{x \mid \exists y, (x,y) \in R\}$                          |
| Range                | $\text{ran}(R) = \{y \mid \exists x, (x,y) \in R\}$                          |
| Codomain             | Declared target set of the relation                                          |
| Reflexive            | $\forall x \in A: (x,x) \in R$                                               |
| Irreflexive          | $\forall x \in A: (x,x) \notin R$                                            |
| Symmetric            | $(x,y) \in R \Rightarrow (y,x) \in R$                                        |
| Antisymmetric        | $(x,y), (y,x) \in R \Rightarrow x=y$                                         |
| Transitive           | $(x,y), (y,z) \in R \Rightarrow (x,z) \in R$                                 |
| Trichotomy           | $x \neq y \Rightarrow (x,y) \in R \text{ or } (y,x) \in R$                   |
| Inverse              | $(x,y) \in R \Leftrightarrow (y,x) \in R^{-1}$                               |
| Composition          | $S \circ R = \{(a,c) \mid \exists b: (a,b) \in R \text{ and } (b,c) \in S\}$ |
| Transitivity theorem | $R \text{ transitive} \Leftrightarrow R \circ R \subseteq R$                 |

## 21. Final Revision Checklist

- ✓ I can explain why  $(a,b)$  and  $(b,a)$  are normally different.
- ✓ I can form  $A \times B$ ,  $B \times A$  and  $A^2$  and use  $|A \times B| = |A||B|$  as a check.
- ✓ I can decide whether a set of pairs is a relation from  $A$  to  $B$ .
- ✓ I can find  $\text{dom}(R)$ ,  $\text{ran}(R)$  and the codomain.
- ✓ I can explain  $R \subseteq A^2$  and the term binary relation.
- ✓ I can test reflexive and irreflexive correctly, including the “neither” case.
- ✓ I can test symmetric and antisymmetric without confusing them.
- ✓ I can systematically test every transitive chain.
- ✓ I can test trichotomy by checking every pair of distinct elements.
- ✓ I can construct a relation with required properties.
- ✓ I can reverse all pairs to find  $R^{-1}$ .
- ✓ I can compute  $S \circ R$  by matching middle coordinates and remembering  $R$  goes first.
- ✓ I can solve combinations such as  $R \circ R$ ,  $R^{-1} \circ R$  and  $R \circ R^{-1}$ .
- ✓ I can prove or explain why  $R$  is transitive iff  $R \circ R \subseteq R$ .

### Bottom line

If you can do every checklist item without looking at the notes, you have covered the complete conceptual and exercise scope of COS1501 Study Unit 5 in the supplied study guide, with additional worked practice.

## 22. Practice Test

- Let  $A = \{1,2\}$  and  $B = \{x,y,z\}$ . Write  $A \times B$  and  $B \times A$ .
- Let  $R = \{(1,a), (2,a), (2,b)\}$  be from  $\{1,2,3\}$  to  $\{a,b,c\}$ . Find  $\text{dom}(R)$ ,  $\text{ran}(R)$ , codomain.
- On  $A = \{1,2,3\}$ , classify  $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$  as reflexive, irreflexive, symmetric and antisymmetric.
- Determine whether  $R = \{(1,2), (2,3), (1,3)\}$  on  $\{1,2,3\}$  is transitive.
- Determine whether  $R = \{(2,1), (3,2)\}$  on  $\{1,2,3\}$  satisfies trichotomy.
- Find the inverse of  $R = \{(a,1), (b,2), (a,2)\}$ .
- Let  $R = \{(1,a), (2,b)\}$  and  $S = \{(a,x), (b,y), (b,z)\}$ . Find  $S \circ R$ .
- Give one relation on  $\{a,b,c\}$  that is simultaneously symmetric and antisymmetric.
- Give one counterexample showing that “not symmetric” does not mean antisymmetric.
- Explain in one sentence why  $R \circ R \subseteq R$  is the same idea as transitivity.

### Practice Test Answers

- $A \times B = \{(1,x), (1,y), (1,z), (2,x), (2,y), (2,z)\}$ ;  $B \times A = \{(x,1), (x,2), (y,1), (y,2), (z,1), (z,2)\}$ .
- $\text{dom}(R) = \{1,2\}$ ;  $\text{ran}(R) = \{a,b\}$ ; codomain =  $\{a,b,c\}$ .

3. Reflexive yes; irreflexive no; symmetric yes; antisymmetric no because (1,2) and (2,1) occur with  $1 \neq 2$ .
4. Yes. The only nontrivial chain  $1 \rightarrow 2 \rightarrow 3$  has required shortcut  $1 \rightarrow 3$ .
5. No. Elements 1 and 3 are incomparable: neither (1,3) nor (3,1) is present.
6.  $R^{-1} = \{(1,a), (2,b), (2,a)\}$ .
7.  $S \circ R = \{(1,x), (2,y), (2,z)\}$ .
8. Identity relation  $\{(a,a), (b,b), (c,c)\}$ .
9. For example  $\{(1,2), (2,1), (2,3)\}$  is not symmetric and also not antisymmetric.
10.  $R \circ R$  contains exactly the endpoints of every two-step R-chain; transitivity requires all those endpoints to already be related by R.

## 23. Unit 5 Vocabulary Glossary

| Term              | Meaning in this unit                                                  |
|-------------------|-----------------------------------------------------------------------|
| Ordered pair      | A two-coordinate object (x,y) in which position matters.              |
| Cartesian product | The set of all possible ordered pairs from two sets.                  |
| Relation          | A subset of a Cartesian product.                                      |
| Domain            | First-coordinate values that actually occur in the relation.          |
| Range             | Second-coordinate values that actually occur.                         |
| Codomain          | The full declared target set.                                         |
| Binary relation   | A relation whose members are ordered pairs.                           |
| Reflexive         | Every element is related to itself.                                   |
| Irreflexive       | No element is related to itself.                                      |
| Symmetric         | Every relation pair has its reverse.                                  |
| Antisymmetric     | Distinct elements cannot be related in both directions.               |
| Transitive        | Every two-step relation chain has the direct shortcut.                |
| Trichotomy        | Every two distinct elements are comparable in at least one direction. |
| Inverse relation  | Relation obtained by reversing all ordered pairs.                     |
| Composition       | Relation formed by joining compatible two-step paths.                 |

Source basis: UNISA COS1501 Study Unit 5 - Relations (pp. 69-82 of the supplied study guide).